Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester& Branch: VI Semester & B.Tech AI&DS
Course Code & Title: CS8391 Embedded Systems and IoT
Name of the Faculty member: Mr.R.Ramasamy
Date: 18.04.2024

Innovative Practice Description

- **Unit / Topic:** Unit V / Applications Development.
- **Course Outcome:** CO5
- **UO Mapping:** 5b
- **Activity Chosen:** Jigsaw
- **Justification:**

Jigsaw is a powerful active learning strategy that encourages students to become responsible for their own learning while also fostering teamwork and cooperation. It is particularly effective for complex topics and subjects that benefit from multiple perspectives and insights. It is often used in educational settings to promote collaborative learning and deeper understanding of complex subject matter.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**
  - **Materials for the Activity:**

The materials for the preparation of the students are shared one week before through LMS Canvas. Including the material, students are asked to refer to the following book and web content also.

Book Title/ Author/ Publisher/ Edition	Page No.
1. Bahga,.Arshdeep, and Vijay Madisetti. *Internet of Things: A hands-on approach*. Vpt, 2014.	~~58~~5-58
Websites	
1. https://bridgera.com/the-role-of-iot-in-smart-city/ 2. https://stefanini.com/en/insights/news/top-5-applications-of-iot-in-building-smart-cities	

- The topic on Smart City was subdivided into five sub categories. The Students are divided into group of five and six students.
- Each group is allotted with a name and it is known as Expert group EG1, EG2, EG3, EG4, EG5 & EG6
- Each expert group is allotted with one topic
  - **Smart Parking**
  - **Smart Lighting**

- **Smart Roads**
- **Structural Health Monitoring**
- **Surveillance**
- **Emergency Response**

- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas. The Expert group members should go to other group and discuss the points to the other members.
- All the members in home group learnt about all the topics through expert group members. Finally, one from each group should summarize the points of the topic Smart City to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)
- Thus the concept of Smart City is known to all by involving the students actively.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO5	H	M	H	M	M	L	L	L	M	L	L	L	M	M	-	M

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 1**	**PO 3**	**PSO 1**
	H	**H**	**M**
Justification for correlation	Students will be able to review and analyse and characterise the Smart City. So, Course outcome is mapped at level H.	By studying the types of Smart City technologies based on a collaborative activity, Jigsaw helps the students present the results as a team, with smooth integration of contributions from all individual efforts. Hence, the outcome is mapped to level M.	The students will be able to identify the suitable sensors for Smart City. Thus, it is mapped at level M.

- **Key Learning:**

Smart Parking:

- Participants learned about the importance of traditional parking systems by integrating technology to optimize space usage, reduce congestion, and enhance user experience.
- They understood the significance of parking operations for municipalities, businesses, and parking lot operators, leading to improved traffic flow, reduced carbon emissions, and increased revenue.

- Discussions highlighted the role of smart parking data empower urban planners to make data-driven decisions for future infrastructure development, ultimately contributing to smarter, more sustainable cities.

Smart Lighting:

- Participants gained insights into the innovative approach to illumination, leveraging advanced technologies like IoT (Internet of Things) and sensors to enhance energy efficiency, convenience, and safety. By incorporating sensors, such as motion detectors or ambient light sensors, smart lighting systems can adjust brightness levels and turn lights on or off automatically based on occupancy or natural light conditions.
- They learned about the impact of remotely controlled and programmed via mobile apps or voice commands, offering users personalized lighting preferences and scheduling options.
- Discussions emphasized the importance to monitor energy usage and performance in real-time, smart lighting solutions not only reduce electricity consumption and operational costs but also contribute to sustainable practices and create more comfortable and secure environments for users.

Smart Roads:

- Participants explored the versatility transformative approach to transportation infrastructure, integrating cutting-edge technologies to enhance safety, efficiency, and sustainability. These roads incorporate a range of intelligent features such as embedded sensors, communication networks, and autonomous vehicle infrastructure.
- They learned about the smart roads facilitate the implementation of connected and autonomous vehicles (CAVs), enabling vehicles to communicate with each other and with the infrastructure, leading to safer and more efficient transportation systems.
- Discussions highlighted the integration of smart roads can incorporate renewable energy sources and energy-harvesting technologies to power lighting and other infrastructure components, contributing to environmental sustainability

Structural Health Monitoring:

- Participants discovered the convenience of critical technique employed in civil engineering and infrastructure management, utilizing advanced sensing technologies to assess the condition and integrity of structures such as buildings, bridges, dams, and pipelines.
- They learned about the integration of real-time data collected from these sensors are analyzed to detect anomalies, assess structural performance, and predict potential failures, enabling early intervention and preventive maintenance measures.
- Discussions focused on the role of SHM enhance safety, prolongs the lifespan of infrastructure assets, and reduces maintenance costs by providing actionable insights into structural health and performance.

Surveillance:

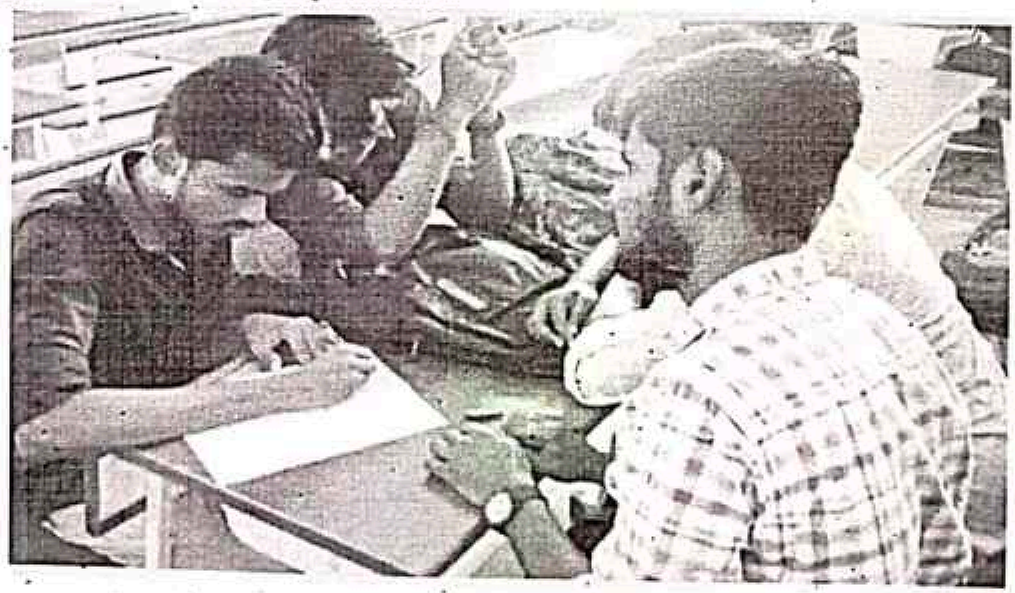
- Participants explored the benefits of behaviors, or information for the purpose of managing, influencing, or protecting individuals, groups, or assets. Utilizing a variety of technologies such as cameras, sensors, and data analytics, surveillance systems are employed across diverse domains including security, law enforcement, and public safety, as well as in commercial and industrial applications.

- They learned about the role of interpretation data to detect anomalies, identify patterns, and mitigate risks in real-time.
- Discussions emphasized the importance of security and situational awareness, it also raises concerns regarding privacy, ethics, and civil liberties, necessitating careful consideration of legal and ethical frameworks to balance security needs with individual rights and freedoms.

Emergency Response

- Participants discovered the convenience of unexpected events such as natural disasters, accidents, or terrorist attacks. It involves a range of activities including preparedness, mitigation, response, and recovery, aimed at protecting lives, property, and the environment.
- They learned about the role of emergency response teams, which may include first responders, law enforcement agencies, medical personnel, and government officials, work together to assess the situation, provide immediate assistance to affected individuals, and implement strategies to restore safety and stability.
- Discussions emphasized the importance of advanced technologies such as communication systems, geographic information systems (GIS), and predictive analytics play a crucial role in enhancing the effectiveness and efficiency of emergency response efforts by facilitating rapid communication, resource allocation, and decision-making in high-pressure situations.

- **Image / Screenshot of the practice:**



Participants provided positive feedback on the effectiveness of the jigsaw activity in enhancing their understanding of IOT based smart city application. The collaborative nature and the opportunity to present and discuss findings were highlighted as valuable components of the activity.

References:

- Bahga, Arshdeep, and Vijay Madisetti. *Internet of Things: A hands-on approach.* Vpt, 2014.
- Zanella, Andrea, et al. "Internet of things for smart cities." *IEEE Internet of Things journal* 1.1 (2014): 22-32.
- Tanwar, Sudeep, Sudhanshu Tyagi, and Sachin Kumar. "The role of internet of things and smart grid for the development of a smart city." *Intelligent Communication and Computational Technologies: Proceedings of Internet of Things for Technological Development, IoT4TD 2017*. Springer Singapore, 2018.

18/04/24

Signature of Faculty Member

4/6/24

HOD

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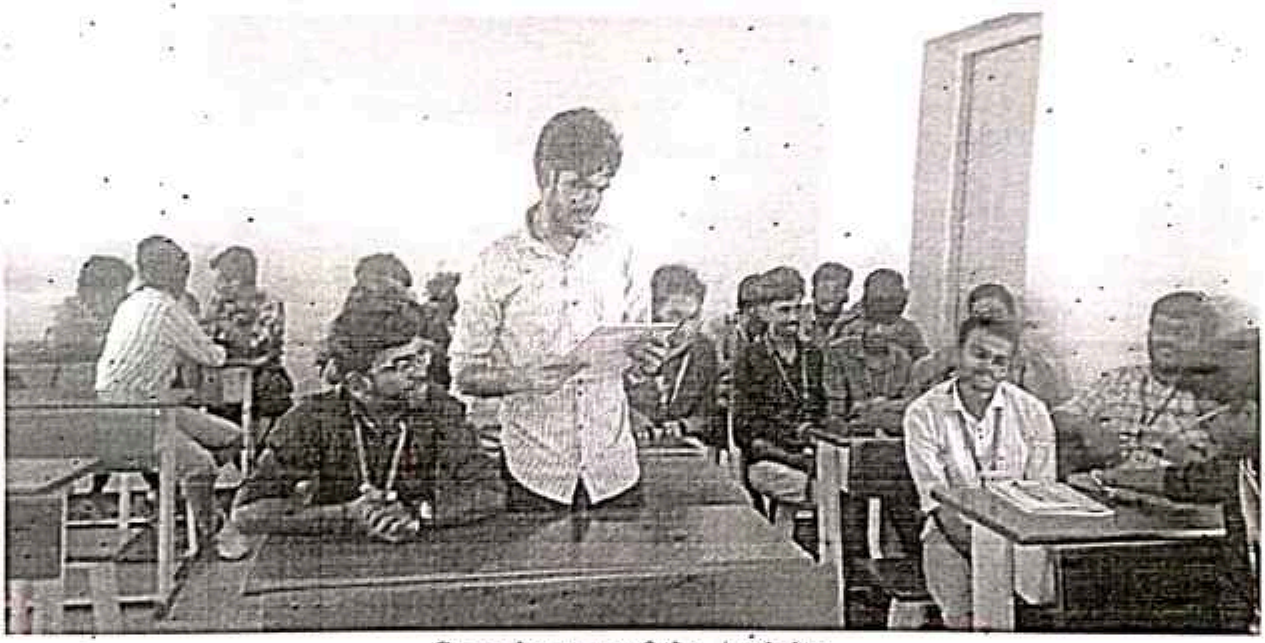
Innovative Practice Description

Unit III: IOT AND ARDUINO PROGRAMMING
Activity: Zero Minute Speech
Topic: IoT Devices Versus Computers
CO 3: Design embedded applications for real-world challenges.
Date: 06.03.2024



Sample snap of the Activity

Justification: A "zero-minute speech" is a powerful tool for conveying essential messages with clarity and precision, making it highly memorable and impactful. Its brevity respects the audience's limited attention span and time, ensuring engagement and focus. This format is particularly valuable in situations with strict time constraints or emergencies, where quick, efficient communication is crucial. Strategically, a concise speech serves as a strong call to action, emphasizing key points without unnecessary details.

06/03/24
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Department of Electronics and Communication Engineering

Academic Year: 2023- 2024 (Even Semester)

Active Learning Practices

Degree, Semester& Branch: VI Semester & B.Tech AI&DS
Course Code & Title: CS8391 Embedded Systems and IoT
Name of the Faculty member: Mr.R.Ramasamy
Date: 02.03.2024

Active Learning Practices Execution

UNIT II EMBEDDED C PROGRAMMING

Activity: Open Book Test
Topic: Priority Based Scheduling Policies

Priority based Scheduling

Lavanya M
Umadevi. M.

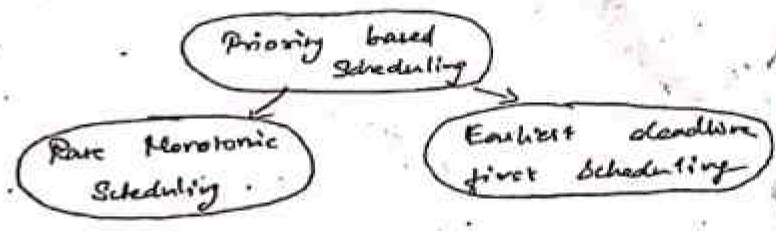
* In Priority Scheduling, an OS kernel determine which process is to be executed next.

* For each task a numerical Priority is assigned.

* The kernel can simply look at the processes and their priorities. It sees which one actually wants to execute and selects the highest priority process that is ready to run.

* This mechanism is flexible and fast.

* It has three types of scheduling namely, Rate Monotonic Scheduling, Earliest deadline first scheduling and optimal dynamic priority scheduling.



Priority based scheduling :

V. Poorna Siva

In priority based scheduling, an OS kernel determines which process is to be executed next. For each task a numerical priority is assigned. The kernel can simply look at the processes and thier priorities.

It sees which one actually wants to execute and selects the highest priority process that is ready to run.

Observation:

Open book tests encourage understanding over memorization, reducing test anxiety and simulating real-world scenarios where individuals access resources to solve problems. These tests promote better study habits, higher-order thinking, and reduce cheating by focusing on how well students use their materials. However, they present challenges in question design, as effective questions must provoke critical thinking rather than simple recall. Students must also manage their time efficiently, balancing the use of resources with answering questions. While open book tests can vary in effectiveness based on students' familiarity with their materials, they ultimately provide a realistic, less stressful assessment environment that fosters a deeper grasp of the subject matter.

27/03/24

Signature of the faculty

6/7/11

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Innovative Practice Description

Unit I: 8-BIT EMBEDDED PROCESSOR
Activity: Think Pair Share
Topic: Timers and Serial Port
CO 1: Understand the internal architecture of embedded processors, including core components such as registers, memory, and execution units.
Date: 15.02.2024

Sample snap of the Activity

Justification: Think-Pair-Share is a collaborative learning strategy that enhances critical thinking and communication by combining individual reflection with group discussion. First, participants independently contemplate a specific question or problem, formulating their own ideas. Next, they pair up to discuss their thoughts, exchanging perspectives and refining their solutions through feedback. Finally, pairs share their conclusions with the larger group, allowing everyone to benefit from the diverse ideas and fostering a comprehensive understanding. This method promotes active participation, improves communication skills, and leads to deeper, collective insights.

15/02/24
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